LONG DOCUMENTS · MULTI-STEP QUESTIONS

More words. Worse answers.

We hid the answers to step-by-step puzzles inside documents that grew longer and longer, then asked three Claude models to solve them. Every model lost accuracy as the document grew. Some lost more than others.

• HAIKU

76% → 47%

Short report → short-book document

Slips first. Loses 29 points. Fast and cheap, but the first to fold once text gets long.

• SONNET

80% → 62%

Short report → short-book document

Strong but slow. Loses 18 points. Hard answers can take 5–6 minutes each.

• OPUS

76% → 63%

Short report → short-book document

The quiet winner. Holds up best on long documents and stays sharp on hard questions — and it's twice as fast as Sonnet.

WHAT WE LEARNED ABOUT LENGTH

Long documents hurt every model. Haiku most.

On a short report, all three score 76–80%. On a short-book-length document, Haiku drops to 47% while Sonnet and Opus stay above 60%. The longer the document, the wider the gap between cheap and premium models.

WHAT WE LEARNED ABOUT DEPTH

Stacked questions break Haiku first.

A simple "what is X?" is easy for everyone. Asking "find X, then use it to find Y, then Z" is harder. By the third step, Haiku starts to slip. By the sixth, it's 15 points behind Sonnet and Opus.

What to do about it →

Six things you can change today, based on what the numbers above are telling you.

1 **Trim what you paste in.**

Haiku loses 30 points of accuracy when documents grow long. Paste only what the model needs — the function, not the whole repo. The paragraph, not the whole PDF.

Example: refactoring a function? Paste the function and its callers, not the whole codebase. Less to read = fewer hallucinations.

2 **Restart long chats.**

Every back-and-forth piles onto the conversation. After ~10 messages, your chat is as long as a research paper — and the model's accuracy is paying the same price as it would on any long document.

Example: stuck on the same bug after 8 messages? Start a new session with a one-paragraph recap of what you've tried.

3 **For tricky multi-step tasks, skip Haiku.**

Once a task chains more than three lookups together ("find this, then use it to find that, then..."), Haiku starts slipping fast. Sonnet and Opus stay sharp through six or more steps.

Example: tracing a bug across 5 files? Use Opus or Sonnet. Asking who wrote a single line? Haiku is fine.

4 **Split big prompts into smaller ones.**

A prompt that asks "find this, then summarize, then translate" is three problems stacked on top of each other. Each extra step costs about 10 points of accuracy. Send three separate prompts.

Example: instead of "extract the requirements, write the spec, then write the tests", do those one at a time and feed each output into the next call.

5 **Match the model to the task shape.**

Short text, simple question → Haiku (fast and cheap). Long document with multi-step questions → Opus (best accuracy, twice as fast as Sonnet). Need the absolute best output and time isn't the worry → Sonnet.

Example: customer FAQ bot → Haiku. Reasoning over a long contract → Opus. Drafting a polished proposal → Sonnet.

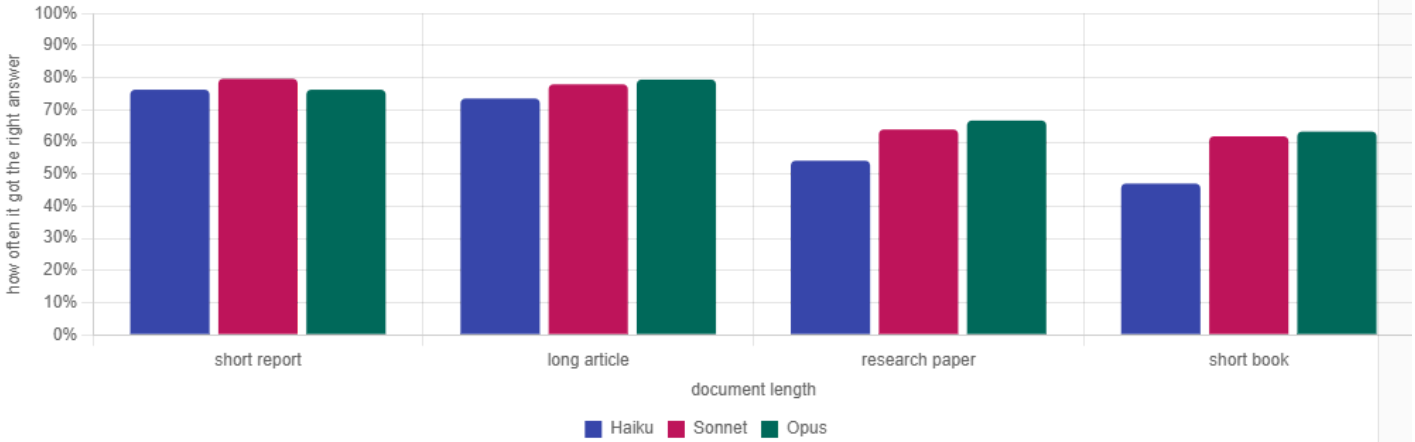
6 **Shrink the prompt before upgrading the model.**

Going from Haiku to Opus is roughly 15× more expensive. Trimming the prompt is free, and often fixes the same problem.

Example: Haiku giving wrong code? Try cutting the context in half before reaching for a bigger model.

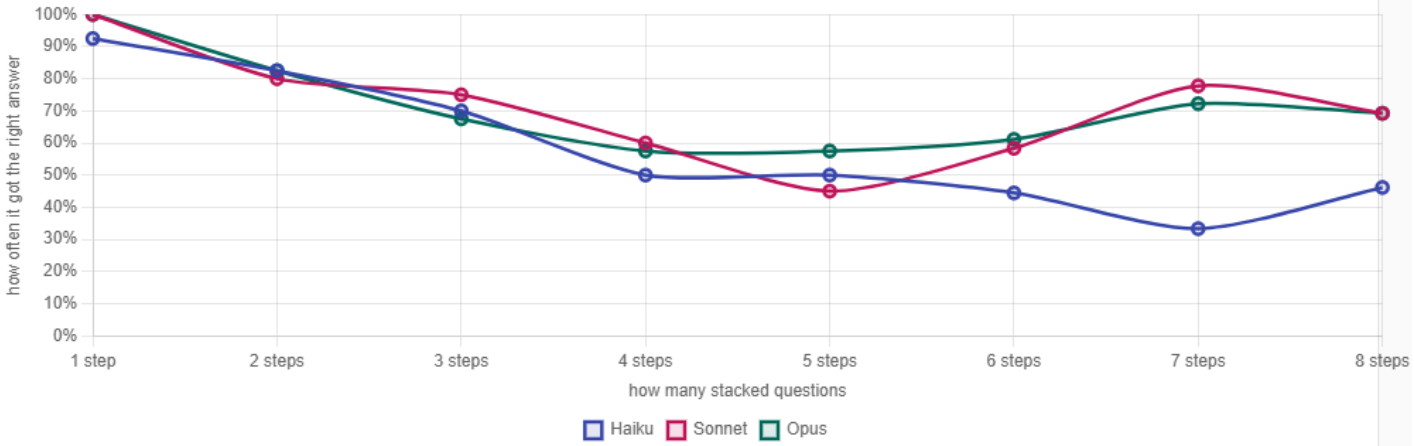
Bigger documents → worse answers

How often each model got the right answer, averaged across all question types, for each document length.



More steps → more mistakes

How often each model got the right answer, averaged across all document lengths, by how many lookups it had to do in a row.



The full picture

Each square shows how often the model got the right answer for one specific (document length × number of stacked questions) combo. Green is good, red is bad. Click a tab to focus on one model.

All models | Haiku | Sonnet | Opus

All three models stacked. Same axes, same color scale.

Haiku 4 document lengths × 8 step counts · 32 combinations

document length	1	2	3	4	5	6	7	8
~1.5k words short report	100% 10/10	80% 8/10	80% 8/10	70% 7/10	60% 6/10	50% 3/6	100% 2/2	100% 1/1
~5.5k words long article	100% 10/10	100% 10/10	70% 7/10	50% 5/10	70% 7/10	70% 7/10	50% 2/4	50% 2/4
~16k words research paper	80% 8/10	80% 8/10	70% 7/10	40% 4/10	50% 5/10	40% 4/10	17% 1/6	33% 2/6
~52k words short book	90% 9/10	70% 7/10	60% 6/10	40% 4/10	20% 2/10	20% 2/10	17% 1/6	50% 1/2

↑ how many lookups in a row · 1 = single question, 8 = eight stacked on top of each other

Sonnet 4 document lengths × 8 step counts · 32 combinations

document length	1	2	3	4	5	6	7	8
~1.5k words short report	100% 10/10	80% 8/10	80% 8/10	80% 8/10	60% 6/10	67% 4/6	100% 2/2	100% 1/1
~5.5k words long article	100% 10/10	80% 8/10	70% 7/10	60% 6/10	80% 8/10	70% 7/10	75% 3/4	100% 4/4
~16k words research paper	100% 10/10	70% 7/10	80% 8/10	60% 6/10	20% 2/10	60% 6/10	83% 5/6	33% 2/6
~52k words short book	100% 10/10	90% 9/10	70% 7/10	40% 4/10	20% 2/10	40% 4/10	67% 4/6	100% 2/2

↑ how many lookups in a row · 1 = single question, 8 = eight stacked on top of each other

Opus 4 document lengths × 8 step counts · 32 combinations

document length	1	2	3	4	5	6	7	8
~1.5k words short report	100% 10/10	70% 7/10	80% 8/10	70% 7/10	60% 6/10	67% 4/6	100% 2/2	100% 1/1
~5.5k words long article	100% 10/10	100% 10/10	70% 7/10	80% 8/10	70% 7/10	60% 6/10	50% 2/4	100% 4/4
~16k words research paper	100% 10/10	80% 8/10	60% 6/10	40% 4/10	60% 6/10	60% 6/10	100% 6/6	33% 2/6
~52k words short book	100% 10/10	80% 8/10	60% 6/10	40% 4/10	40% 4/10	60% 6/10	50% 3/6	100% 2/2

↑ how many lookups in a row · 1 = single question, 8 = eight stacked on top of each other

0% 100% Empty cell = not tested at this combination

HOW TO READ THIS

- **Document length** is how much surrounding text wraps the actual answer. From shortest to longest: a short report (~1.5k words), a long article (~5.5k), a research paper (~16k), and a short book (~52k).
- **Stacked questions** means how many lookups the model has to do in a row. Step 1 is "Who is X?". Step 4 is "Find X, use X to find Y, use Y to find Z, then answer one more thing." Each step builds on the previous answer.
- **Some squares are nearly empty.** A few combinations could only ask 1 or 2 questions. A 100% there could be luck — don't read those as recoveries.

- **Time matters as much as accuracy** for production use. Sonnet hits 5–6 minutes per answer on the hardest cells. Opus stays around 2 minutes for similar accuracy. Haiku is fastest but trades it for a lower ceiling.